

Xorry 1

You are given an integer X . Two integers (A, B) are said to be an Xorry pair, if the following conditions are satisfied:

- $A \oplus B = X$
- $0 \leq A \leq B \leq X$
- Among all the pairs which satisfy the above two conditions, $(B - A)$ is as small as possible

For example, suppose $X = 6$.

Then, the pair $(0, 6)$ satisfies the first two conditions. But, the pair $(2, 4)$ also satisfies the first two conditions. And since, $(6 - 0)$ is larger than $(4 - 2)$, the pair $(0, 6)$ does not have the smallest possible difference, and so, it is not an Xorry pair.

You can verify that the pairs $(2, 4)$ and $(3, 5)$ are the only two Xorry pairs for $X = 6$. This is explained further in the sample explanation below.

Output any one Xorry pair for a given X .

Note that $\oplus$ refers to [bitwise XOR](#).

Input Format

- The first line of input will contain a single integer T , denoting the number of test cases.
- Each test case consists of a single line which contains a single integer, X .

Output Format

For each test case, output on a new line, two space separated integers, A and B ,